

Large Language Models

On Hallucinations and Prompt Engineering

Hallucination

- When a language model generates text that sounds plausible, authoritative even, but is factually incorrect or entirely made up.
- Who won the world cup in 2040?
- AI: US won the 2040 World Cup, 3-2
- What new discoveries from the James Webb Space Telescope can I tell my 9 year old about?
- Bard:.... took the very first pictures of a planet outside of our own solar system

The Misnomer of "Hallucination"

- Implies a failure to be factual
- Unnecessary anthropomorphism—AI doesn't "perceive" or think
- It is a Large Language Model - not a Large Knowledge Model
- Moreover, training data was never labeled fact or fiction
 - Harry gripped his broomstick and soared into the sky
 - Harry gripped his broomstick and swept the ground

Prompts

- Prompt: the text you begin your conversation with. Most often the question you ask.
- LLMs start by ‘predicting’ the next word and keeps doing this until it predicts a *end of sequence (EOS)*
- The issue is some prompts yield better answers!
 - What’s the best way to run?
 - Explain the best techniques for running a 5K race for beginners.

Prompt Engineering

- Art and science of designing inputs (prompts) to guide AI models toward producing useful, accurate, and relevant outputs.
- Mostly common sense and realizing that the AI does not know who you are, what you are going through, what you like etc.
- It is more about avoiding bad prompts!

Bad Prompts!

- Vagueness : Tell me about science?
- Ambiguity : Explain the fall.
- Overcomplexity: Write a story about a dog who's a detective and also summarize WWII and list 10 recipes.
- Lack of Context: Fix this.
- Unrealistic Expectations: Predict the exact weather in NY on March 26, 2050
- Missing Constraints: Explain gravity.
- Tone or Style Mismatch: Write a letter to my boss
- Redundancy: Tell me about cats, like what cats are, feline animals, you know, cats.
- Assumption of Prior Knowledge: What about the other one?
- Contradictory Instructions: Write a short, detailed essay about space

Prompt refining

- Smarter LLMs + prompt refinement = less need for manual crafting?
- Seeking clarifications: LLMs are being designed to ask questions when they are not certain what you want
- Having Memory